

LECH'S CONJECTURE IN EQUAL CHARACTERISTIC

SRIKANTH B. IYENGAR, LINQUAN MA, MARK WALKER

ABSTRACT. This article is a digestion of a proof of Lech's conjecture in equal characteristic, found by Alex Gerko using OpenAI's ChatGPT Astra. We have made the original, entirely AI written, paper available for those interested in seeing it at Walker's website; go to <https://mark-walker-22.github.io> or just click [here](#).

Disclaimer. First of all, we would like to make clear that we make no claim to owning the ideas found in this paper. Rather, our aim is to make the original AI-generated proofs more accessible to a human audience by improving the exposition, supplying additional context and references, and, in some occasions, refining or simplifying the original arguments.

We also wish to emphasize that this is an extremely preliminary draft – much rougher than a version we would be comfortable circulating under ordinary circumstances. We hope to release a more polished and detailed preprint in the near future. We believe that the proofs are correct, but our understanding and the level of confidence are perhaps best compared to those of a referee vouching for the validity of a paper, rather than those of authors who have fully internalized and developed every aspect of the arguments themselves.

We are releasing this draft now in the hope that the mathematical community will benefit from seeing these ideas right away, and also that others may help us create a better document. In particular, we do not yet fully understand which, if any, previously existing ideas (beyond those already cited here) may have been drawn upon by the AI in producing the proofs in this paper. We have made some effort to track these ideas down, but our search may well be incomplete. We would therefore be grateful to hear from anyone who recognizes an argument, technique, or idea that has appeared before and is not currently credited appropriately.

The proofs in [Section 3](#) are quite different from the original ones in the document provided to us by Alex Gerko. We arrived at the current proofs using additional prompts to ChatGPT Astra, thus these arguments should also be considered to be due to AI, and are not ours.

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1. INTRODUCTION

The goal is to explain the proof of the following theorem, which settles the long standing Lech's conjecture [[Lec60](#)] in commutative algebra in equal characteristic.

Theorem 1.1. *Let $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ be a flat local extension of Noetherian local rings. Suppose R contains a field. Then $e(R) \leq e(S)$.*

In our previous works [Ma17, Section 5] and [IMW22, Proposition 8.3], we have reduced [Theorem 1.1](#) to proving the following lower bound on Dutta multiplicity (see [Section 2](#) for a definition of χ_∞ in characteristic $p > 0$).

Theorem 1.2. *Let $(R, \mathfrak{m})$ be a Noetherian complete local ring of characteristic $p > 0$ and dimension d . Let $F_\bullet := 0 \rightarrow F_d \rightarrow \cdots \rightarrow F_1 \rightarrow F_0 \rightarrow 0$ be a perfect complex over R of finite length homology. Then*

$$\chi_\infty(F_\bullet) \geq e(R).$$

The proof of [Theorem 1.2](#) combines characteristic $p > 0$ method together with the following key result that bounds the length of a finite length module over a regular local ring when its minimal free resolution is “deep” in an appropriate sense.

Theorem 1.3. *Let $(A, \mathfrak{m}_A)$ be a regular local ring of equal characteristic of dimension d and let M be a finite length A -module. Fix an integer $N > 0$. Suppose there exists a finite free resolution $F_\bullet$ of M over A such that with a chosen basis of each F_i , there are at most C columns among all the differentials so that the entries in those columns are not necessarily contained in $\mathfrak{m}_A^N$. Then*

$$\ell_A(M) \geq (\nu_A(M) - 3C)N^d.$$

In particular, if all the differentials in the minimal free resolution $F_\bullet$ of M over A have entries in $\mathfrak{m}_A^N$ (i.e., $\partial(F_\bullet) \subseteq \mathfrak{m}_A^N F_\bullet$), then

$$\ell_A(M) \geq \nu_A(M)N^d.$$

The proof of [Theorem 1.3](#) is inspired from Boij–Söderberg theory [ES09]. In fact, if A is a standard graded polynomial ring over a field and M is graded, then the proof of [Theorem 1.3](#) uses ideas from the positivity mechanism underlying Boij–Söderberg theory. Passing from the graded case to the local case relies on a very delicate (and original, at least to the authors) homogenization technique.

This article is organized as follows: in [Section 2](#) we prove [Theorem 1.2](#) assuming [Theorem 1.3](#); in [Section 3](#) we prove [Theorem 1.3](#) in the graded case and in [Section 4](#) we prove [Theorem 1.3](#) in general by reducing it to the graded case.

2. PROOF OF [THEOREM 1.2](#)

In this section we prove the lower bound on Dutta multiplicity assuming [Theorem 1.3](#). We first recall that, if $(R, \mathfrak{m})$ is a Noetherian local ring of characteristic $p > 0$ and $G_\bullet$ is a perfect complex supported only at $\{\mathfrak{m}\}$ (equivalently, $G_\bullet$ has finite length homology, i.e., $\ell_R(H_i(G_\bullet)) < \infty$ for all i), then the Dutta multiplicity of $G_\bullet$ is defined as

$$\chi_\infty(G_\bullet) := \lim_{e \rightarrow \infty} \frac{\chi(\mathcal{F}_R^e(G_\bullet))}{p^{ed}}$$

where $\mathcal{F}_R^e(-)$ denotes Peskine–Szpiro’s Frobenius functor. More concretely, $\mathcal{F}_R^e(G_\bullet)$ is simply the perfect complex obtained from G by replacing all entries in the differentials by their p^e -th powers. It is easy to see that if $(R, \mathfrak{m}) \rightarrow (R', \mathfrak{m}')$ is any flat local extension with $\mathfrak{m}' = \mathfrak{m}R'$, then $\chi_\infty(G_\bullet) = \chi_\infty(G'_\bullet)$ where $G'_\bullet = G_\bullet \otimes_R^{\mathbb{L}} R'$.

We now prove [Theorem 1.2](#) when R is Cohen–Macaulay – it turns out that, remarkably, this case follows almost immediately from [Theorem 1.3](#).

Proof of Theorem 1.2 assuming R is Cohen–Macaulay. Without loss of generality, we may assume that the residue field of R is infinite. Let k be a coefficient field of R , $x_1, \dots, x_d$ be a minimal reduction of $\mathfrak{m}$, and $(A, \mathfrak{m}_A) := k[[x_1, \dots, x_d]]$. Since R is Cohen–Macaulay, R is a finite free A -module. Since $x_1, \dots, x_d$ is a minimal reduction of $\mathfrak{m}$, we have

$$\nu_A(R) = \text{rank}_A(R) = e(R).$$

By the acyclic lemma [PS73, Lemme (I.8)], we may assume that $F_\bullet$ is a minimal free resolution of a finite length R -module M . Moreover, by [PS73, Théorème (I.7)], we know that $\mathcal{F}_R^e(F_\bullet)$ is a minimal finite free resolution of $\mathcal{F}_R^e(M)$. By restriction of scalars and using that R is finite free over A , we can view $\mathcal{F}_R^e(F_\bullet)$ as a finite free resolution of $\mathcal{F}_R^e(M)$ over A .

Next we note that all the differentials in $\mathcal{F}_R^e(F_\bullet)$ are contained in $\mathfrak{m}_R^{[p^e]}$ and we have

$$\mathfrak{m}_R^{[p^e]} \subseteq \mathfrak{m}_R^{p^e} \subseteq \mathfrak{m}_A^{p^e - c} R$$

where c is a constant independent of e (the second inclusion uses the fact that $\mathfrak{m}_A R$ is a reduction of $\mathfrak{m}_R$). It follows that $\mathcal{F}_R^e(F_\bullet)$, when viewed as a perfect complex over A , has all entries of its differentials in $\mathfrak{m}_A^{p^e - c}$. Hence we can apply Theorem 1.3 with $N = p^e - c$ and $C = 0$ to obtain that

$$(2.0.1) \quad \chi(\mathcal{F}_R^e(F_\bullet)) = \ell_R(\mathcal{F}_R^e(M)) = \ell_A(\mathcal{F}_R^e(M)) \geq \nu_A(\mathcal{F}_R^e(M))(p^e - c)^d.$$

Finally, we note that, as the entries in the first differential of $\mathcal{F}_R^e(F_\bullet)$ are contained in $\mathfrak{m}_A$, $\nu_A(\mathcal{F}_R^e(M))$ is simply the 0-th Betti number of $F_\bullet$ (viewed as a finite free A -complex). Thus

$$\nu_A(\mathcal{F}_R^e(M)) = \nu_R(M) \cdot \nu_A(R) = \nu_R(M) \cdot e(R).$$

Thus after dividing (2.0.1) by p^{ed} and taking the limit as $e \rightarrow \infty$, we find that

$$\chi_\infty(F_\bullet) \geq \nu_R(M) \cdot e(R) \geq e(R). \quad \square$$

In the rest of this section, we prove the general case of Theorem 1.2. In fact, we will prove the slightly stronger statement $\chi_\infty(F_\bullet) \geq \nu_R(H_0(F_\bullet)) \cdot e(R)$, just as in the Cohen–Macaulay case. The argument is more involved and crucially relies on characteristic $p > 0$ methods. We start with a lemma.

Lemma 2.1. *Let $(A, \mathfrak{m}_A) \rightarrow (R, \mathfrak{m})$ be a finite extension of Noetherian local domains. Fix an integer $d > 0$. Then there exists a nonzero element $h \in A$ such that for every perfect complex $F_\bullet$ over R with length at most d , there exists a perfect complex $G_\bullet$ over A and a map $G_\bullet \rightarrow F_\bullet$ in $D(A)$ so that h annihilates $\text{cone}(G_\bullet \rightarrow F_\bullet)$ in $D(A)$.*

Furthermore, when $\mathfrak{m}$ is integral over $\mathfrak{m}_A R$, there is an integer c such that if $\partial(F_\bullet) \subseteq \mathfrak{m}^N F_\bullet$ holds for some integer N , then the complex $G_\bullet$ can be chosen with $\partial(G_\bullet) \subseteq \mathfrak{m}_A^{N-c} G_\bullet$.

Proof. Since R is a domain finite over A , we have an injection $f: A^r \rightarrow R$ whose cokernel is torsion over A , where $r = \text{rank}_A(R)$. Let $0 \neq a \in A$ that annihilates the cokernel, we obtain $g: R \rightarrow A^r$ so that $f \circ g$ and $g \circ f$ are both multiplication by a . We write $F_\bullet$ as

$$0 \rightarrow R^{n_d} \rightarrow \dots \rightarrow R^{n_1} \rightarrow R^{n_0} \rightarrow 0.$$

Consider the following diagram with the top row defining $G_\bullet$ and the map between the rows defining $\tilde{f}: G_\bullet \rightarrow F_\bullet$ and $\tilde{g}: F_\bullet \rightarrow G_\bullet$.

$$\begin{array}{ccccccc}
0 & \longrightarrow & A^{rn_d} & \xrightarrow{g \circ \partial_d \circ f} & \cdots & \xrightarrow{g \circ \partial_2 \circ f} & A^{rn_1} & \xrightarrow{g \circ \partial_1 \circ f} & A^{rn_0} & \longrightarrow & 0 \\
& & \downarrow a^d \cdot f & & & & \downarrow a \cdot f & & \downarrow f & & \\
0 & \longrightarrow & R^{n_d} & \xrightarrow{\partial_d} & \cdots & \xrightarrow{\partial_2} & R^{n_1} & \xrightarrow{\partial_1} & R^{n_0} & \longrightarrow & 0 \\
& & \downarrow g & & & & \downarrow a^{d-1} \cdot g & & \downarrow a^d \cdot g & & \\
0 & \longrightarrow & A^{rn_d} & \xrightarrow{g \circ \partial_d \circ f} & \cdots & \xrightarrow{g \circ \partial_2 \circ f} & A^{rn_1} & \xrightarrow{g \circ \partial_1 \circ f} & A^{rn_0} & \longrightarrow & 0 \\
& & \downarrow a^d \cdot f & & & & \downarrow a \cdot f & & \downarrow f & & \\
0 & \longrightarrow & R^{n_d} & \xrightarrow{\partial_d} & \cdots & \xrightarrow{\partial_2} & R^{n_1} & \xrightarrow{\partial_1} & R^{n_0} & \longrightarrow & 0
\end{array}$$

$\cdot a^{d+1}$
 $\cdot a^{d+1}$

It is straightforward to check that $G_\bullet$ is a perfect complex over A , and that $\tilde{f} \circ \tilde{g}$ and $\tilde{g} \circ \tilde{f}$ are both multiplication by $h := a^{d+1}$. Examining the commutative diagram

$$\begin{array}{ccccccc}
G_\bullet & \xrightarrow{\tilde{f}} & F_\bullet & \longrightarrow & \text{cone}(\tilde{f}) & \xrightarrow{+1} & \\
\downarrow \cdot h & & \downarrow \tilde{g} & & \downarrow & & \\
G_\bullet & \xrightarrow{\text{id}} & G_\bullet & \longrightarrow & 0 & & \\
\downarrow \text{id} & & \downarrow \tilde{f} & & \downarrow & & \\
G_\bullet & \xrightarrow{\tilde{f}} & F_\bullet & \longrightarrow & \text{cone}(\tilde{f}) & \xrightarrow{+1} &
\end{array}$$

$\cdot h$

shows that multiplication by h annihilates $\text{cone}(\tilde{f})$, as it factors through $0 \in D(A)$. The last statement is clear from the construction of the differentials in $G_\bullet$, since there exists $c \in \mathbb{N}$ such that $\mathfrak{m}^N \subseteq \mathfrak{m}_A^{N-c} R$ for all N by the assumption that $\mathfrak{m}$ is integral over $\mathfrak{m}_A R$. $\square$

Before proving [Theorem 1.2](#), we introduce some notations. For a perfect complex $G_\bullet$ over a Noetherian local ring $(R, \mathfrak{m})$, we let $\beta_R(G_\bullet)$ denote its total Betti numbers, i.e.,

$$\beta_R(G_\bullet) = \sum_i \ell_R(H_i(G_\bullet \otimes_R^\mathbb{L} R/\mathfrak{m})).$$

Note that if $G_\bullet^1 \rightarrow G_\bullet^2 \rightarrow G_\bullet^3 \xrightarrow{+1}$ is an exact triangle of perfect complexes over R , then it is easy to see that

$$\beta_R(G_\bullet^i) \leq \beta_R(G_\bullet^j) + \beta_R(G_\bullet^k)$$

for all $\{i, j, k\} = \{1, 2, 3\}$. If A is a regular local ring, then every finitely generated A -module M is isomorphic to a perfect complex in $D(A)$ and $\beta_A(M)$ is simply the sum of the Betti numbers in a minimal free resolution of M .

Theorem 2.2. *Let $(R, \mathfrak{m})$ be a Noetherian complete local ring of characteristic $p > 0$ and dimension d . Let $F_\bullet := 0 \rightarrow F_d \rightarrow \cdots \rightarrow F_1 \rightarrow F_0 \rightarrow 0$ be a perfect complex over R of finite length homology. Then*

$$\chi_\infty(F_\bullet) \geq \nu_R(H_0(F_\bullet)) \cdot e(R).$$

Proof. By the associativity formula for the Hilbert–Samuel and the Dutta multiplicity,¹ we have that

$$\chi_\infty(F_\bullet) = \sum_{\dim(R/P)=d} \ell(R_P) \chi_\infty(F_\bullet \otimes_R^{\mathbb{L}} R/P), \quad \text{and} \quad e(R) = \sum_{\dim(R/P)=d} \ell(R_P) e(R/P).$$

Therefore, as $\nu_R(H_0(F_\bullet)) = \nu_R(H_0(F_\bullet \otimes_R^{\mathbb{L}} R/P))$ for every P , without loss of generality we may assume that $(R, \mathfrak{m})$ is a Noetherian complete local domain with an infinite residue field. Let k be a coefficient field of R , $x_1, \dots, x_d$ be a minimal reduction of $\mathfrak{m}$, and $(A, \mathfrak{m}_A) := k[[x_1, \dots, x_d]]$. Since $x_1, \dots, x_d$ is a minimal reduction of $\mathfrak{m}$, we have

$$\text{rank}_A(R) = e(\mathfrak{m}_A, R) = e(R).$$

For each $e \in \mathbb{N}$, set $q := p^e$ and consider the perfect complex $F_\bullet^q := \mathcal{F}_R^e(F_\bullet)$ over R . For each $F_\bullet^q$ we construct a perfect complex $G_\bullet^q$ over A as in [Lemma 2.1](#). In particular, there exists $0 \neq h \in A$ that annihilates $\text{cone}(G_\bullet^q \rightarrow F_\bullet^q)$ for each q , and that the entries in all the differentials of $G_\bullet^q$ are contained in $\mathfrak{m}_A^{q-c}$ (we caution the readers that $G_\bullet^q$ may not have finite length homology). It is well known that (see [\[Rob89\]](#) or [\[BHM26, Theorem 5.2\]](#))

$$\ell_R(H_i(F_\bullet^q)) = \ell_R(H_i(\mathcal{F}_R^e(F_\bullet))) = o(q^d) \text{ for all } i \geq 1.$$

In what follows, we will need an auxiliary sequence of integers $\{t_q\}_q$ such that

$$t_q \rightarrow \infty, \quad t_q = o(q), \quad \text{and} \quad \lim_{q \rightarrow \infty} \frac{\ell_R(H_i(F_\bullet^q))}{t_q^d} = 0 \quad \text{for all } i \geq 1.$$

Such sequence can be constructed, for example by taking

$$t_q := \lfloor (q^d \cdot \sup_i \{\ell_R(H_i(F_\bullet^q)) + 1\})^{\frac{1}{2d}} \rfloor.$$

We next set $M^q := H_0(F_\bullet^q) = H_0(\mathcal{F}_R^e(F_\bullet))$. We have natural maps in $D(A)$:

$$G_\bullet^q \rightarrow F_\bullet^q \rightarrow M^q.$$

Let $Q_1^q = \text{cone}(G_\bullet^q \rightarrow F_\bullet^q)$, $Q_2^q = \text{cone}(F_\bullet^q \rightarrow M^q)$, and $Q^q = \text{cone}(G_\bullet^q \rightarrow M^q)$. By the octahedra axiom, we have an exact triangle:

$$(2.2.1) \quad Q_1^q \rightarrow Q^q \rightarrow Q_2^q \xrightarrow{+1}.$$

Consider the following endomorphism on A (the k -linear t_q -th “Frobenius”):

$$\sigma_{t_q} : A \rightarrow A \quad x_i \mapsto x_i^{t_q}.$$

For any $K \in D(A)$, we write $\sigma_{t_q*} K$ for the restriction of scalars along σ_{t_q} .

Claim 2.3. *We have $\beta_A(\sigma_{t_q*} Q^q) = o(t_q^d)$ as $q \rightarrow \infty$.*

¹The associativity formula for Dutta multiplicity is clear if one uses the characteristic-free definition via localized Chern characters, see [\[Rob89, Rob98\]](#). In our specific characteristic $p > 0$ scenario (the length of $F_\bullet$ equals $d = \dim(R)$), one can also deduce it by observing that the higher terms in the definition vanish asymptotically (for example by [\[BHM26, Theorem 5.2\]](#)), and the degree 0 term, which is a variant of the Hilbert–Kunz multiplicity, satisfies the associativity formula by [\[Sei89\]](#).

Proof of Claim. By the exact triangle (2.2.1), it suffices to show

$$\beta_A(\sigma_{t_q*}Q_1^q) = o(t_q^d) = \beta_A(\sigma_{t_q*}Q_2^q).$$

For the second equality, note that since $Q_2^q = \text{cone}(F_\bullet^q \rightarrow M^q)$ and σ_{t_q} is k -linear (and thus $\sigma_{t_q*}(-)$ does not change length), we know that $\ell(H_i(\sigma_{t_q*}Q_2^q)) = \ell(H_i(Q_2^q)) = o(t_q^d)$ for all i by our choice of t_q . Hence $\beta_A(\sigma_{t_q*}Q_2^q) \leq (\sum_i \ell(H_i(\sigma_{t_q*}Q_2^q))) \cdot \beta_A(k) = o(t_q^d)$.

Now for the first equality, since $\sigma_{t_q*}(-)$ preserves length, we know that

$$\ell_A(H_i(\sigma_{t_q*}Q_1^q \otimes_A^\mathbb{L} k)) = \ell_A(H_i(Q_1^q \otimes_A^\mathbb{L} A/(x_1^{t_q}, \dots, x_d^{t_q})))$$

for all i . Now $H_i(Q_1^q \otimes_A^\mathbb{L} A/(x_1^{t_q}, \dots, x_d^{t_q}))$ is the homology of

$$(A/(x_1^{t_q}, \dots, x_d^{t_q}))^{b_{i+1}} \xrightarrow{\gamma} (A/(x_1^{t_q}, \dots, x_d^{t_q}))^{b_i} \rightarrow (A/(x_1^{t_q}, \dots, x_d^{t_q}))^{b_{i-1}}.$$

Since $0 \neq h \in A$ annihilates $Q_1^q \in D(A)$, we know that h annihilates $H_i(Q_1^q \otimes_A^\mathbb{L} A/(x_1^{t_q}, \dots, x_d^{t_q}))$ for all i . It follows that $H_i(Q_1^q \otimes_A^\mathbb{L} A/(x_1^{t_q}, \dots, x_d^{t_q}))$ is a submodule of $(\text{Image } \gamma : h) / \text{Image } \gamma$. By the exact sequence of finite length A -modules

$$0 \rightarrow \frac{\text{Image } \gamma : h}{\text{Image } \gamma} \rightarrow \frac{(A/(x_1^{t_q}, \dots, x_d^{t_q}))^{b_i}}{\text{Image } \gamma} \xrightarrow{h} \frac{(A/(x_1^{t_q}, \dots, x_d^{t_q}))^{b_i}}{\text{Image } \gamma} \rightarrow \text{coker} \rightarrow 0,$$

we thus find that

$$\ell_A(H_i(Q_1^q \otimes_A^\mathbb{L} A/(x_1^{t_q}, \dots, x_d^{t_q}))) \leq \ell_A(\text{coker}) \leq b_i \cdot \ell_A(A/(x_1^{t_q}, \dots, x_d^{t_q}, h)).$$

Putting these together we have

$$(2.3.1) \quad \beta_A(\sigma_{t_q*}Q_1^q) = \sum_i \ell_A(H_i(Q_1^q \otimes_A^\mathbb{L} A/(x_1^{t_q}, \dots, x_d^{t_q}))) \leq (\sum_i b_i) \cdot \ell_A(A/(x_1^{t_q}, \dots, x_d^{t_q}, h)).$$

Finally, we know that

$$\sum_i b_i = \beta_A(Q_1^q) \leq \beta_A(G_\bullet^q) + \beta_A(F_\bullet^q) \leq \text{rank}_A(R) \cdot \beta_R(F_\bullet) + \beta_R(F_\bullet) \cdot \beta_A(k).$$

This follows from the construction of $G_\bullet^q$ as in Lemma 2.1 since $F_\bullet^q = \mathcal{F}_R^e(F_\bullet)$ (in particular, $F_\bullet^q$ has the same Betti numbers as $F_\bullet$ over R). In particular, $\sum_i b_i$ is bounded by a constant that does not depend on q . Since $\ell_A(A/(x_1^{t_q}, \dots, x_d^{t_q}, h)) = o(t_q^d)$, we obtain from (2.3.1) that $\beta_A(\sigma_{t_q*}Q_1^q) = o(t_q^d)$. This completes the proof of the claim. $\square$

We now continue the proof of Theorem 1.2. We would like to apply Theorem 1.3 to $\sigma_{t_q*}M^q$. Let $P_\bullet^q$ be a free resolution of $\sigma_{t_q*}M^q$ over A . It sits in an exact triangle in $D(A)$:

$$\sigma_{t_q*}G_\bullet^q \rightarrow P_\bullet^q \rightarrow \sigma_{t_q*}Q^q \xrightarrow{+1}.$$

From Lemma 2.1 we know that the entries in all the differentials of $G_\bullet^q$ are contained in $\mathfrak{m}_A^{q-c}$, which implies that the entries in all the differentials of $\sigma_{t_q*}G_\bullet^q$ are contained in $\mathfrak{m}_A^{\lfloor \frac{q-c}{t_q} - d \rfloor}$. This is because

$$(x_1^{1/t_q}, \dots, x_d^{1/t_q})^{q-c} \subseteq (x_1, \dots, x_d)^{\lfloor \frac{q-c}{t_q} - d \rfloor}$$

since if $\sum_i a_i = q - c$, then $\sum_i \lfloor \frac{a_i}{t_q} \rfloor \geq \sum_i \frac{a_i}{t_q} - d \geq \lfloor \frac{q-c}{t_q} - d \rfloor$. Note that $P_\bullet^q$ is isomorphic to $\text{cone}(\sigma_{t_q*}Q^q[-1] \rightarrow \sigma_{t_q*}G_\bullet^q)$ in $D(A)$, thus by analyzing the differentials in the mapping cone and replacing $P_\bullet^q$ by this mapping cone (and trimming it to a minimal free resolution if necessary), we may assume that the entries in all the differentials in $P_\bullet^q$ (with a chosen basis

of P_i^q) are contained in $\mathfrak{m}_A^{\lfloor \frac{q-c}{t_q} - d \rfloor}$, except for at most $\beta_A(\sigma_{t_q*}Q^q)$ many columns. Applying [Theorem 1.3](#) to $\sigma_{t_q*}M^q$, $N = \lfloor \frac{q-c}{t_q} - d \rfloor$, and $C = \beta_A(\sigma_{t_q*}Q^q)$, we obtain that

$$\ell_A(M^q) = \ell_A(\sigma_{t_q*}M^q) \geq (\nu_A(\sigma_{t_q*}M^q) - 3\beta_A(\sigma_{t_q*}Q^q)) \cdot \left(\frac{q-c}{t_q} - d - 1\right)^d.$$

After dividing by q^d , we have that

$$(2.3.2) \quad \frac{\ell_R(H_0(\mathcal{F}_R^e(F_\bullet)))}{p^{ed}} = \frac{\ell_A(M^q)}{q^d} \geq \frac{\nu_A(\sigma_{t_q*}M^q) \cdot (q-c-t_q(d+1))^d}{q^d t_q^d} - \frac{3\beta_A(\sigma_{t_q*}Q^q) \cdot (q-c-t_q(d+1))^d}{q^d t_q^d}.$$

Since $t_q = o(q)$ and $\beta_A(\sigma_{t_q*}Q^q) = o(t_q^d)$ by [Claim 2.3](#), the second factor above tends to 0 when $q \rightarrow \infty$. As for the first factor, we note that $\sigma_{t_q*}M^q$ sits in an exact sequence:

$$\sigma_{t_q*}R^{n_1} \xrightarrow{\sigma_{t_q*}\alpha} \sigma_{t_q*}R^{n_0} \rightarrow \sigma_{t_q*}M^q \rightarrow 0.$$

For $q \gg 0$, we know that $\text{Image}(\alpha) \subseteq \mathfrak{m}^q R^{n_0} \subseteq \mathfrak{m}_A^{q-c} R^{n_0}$. Thus by the same argument as above, we have

$$\text{Image}(\sigma_{t_q*}\alpha) \subseteq \mathfrak{m}_A^{\lfloor \frac{q-c}{t_q} - d \rfloor} \sigma_{t_q*}R^{n_0}.$$

In particular, when $q \rightarrow \infty$, as $t_q = o(q)$ we have

$$\nu_A(\sigma_{t_q*}M^q) = \nu_A(\sigma_{t_q*}R^{n_0}) \geq \text{rank}_A(\sigma_{t_q*}R^{n_0}) = t_q^d \text{rank}_A(R^{n_0}) = t_q^d n_0 \text{rank}_A(R) = t_q^d \nu_R(M^q) e(R).$$

Thus after taking limit as $e \rightarrow \infty$ in [\(2.3.2\)](#), we obtain

$$\chi_\infty(F_\bullet) = \lim_{e \rightarrow \infty} \frac{\ell_R(H_0(\mathcal{F}_R^e(F_\bullet)))}{p^{ed}} \geq \nu_R(M^q) \cdot e(R) = \nu_R(H_0(F_\bullet)) \cdot e(R)$$

where the first equality also uses [\[Rob89\]](#) (see [\[BHM26, Theorem 5.2\]](#)). This completes the proof of the theorem and hence the proof of [Theorem 1.2](#). $\square$

3. PROOF OF [THEOREM 1.3](#): THE GRADED CASE

The goal of this section is to prove [Theorem 3.2](#), which is the graded version of [Theorem 1.3](#). We start by introducing some notations throughout.

Setting 3.1. Let $S = k[x_1, \dots, x_d]$ be a standard graded polynomial ring over a field k and let $\mathfrak{m} = (x_1, \dots, x_d)$. We assume $d \geq 2$.² Let M be a graded S -module of finite length and

$$F_\bullet : 0 \longrightarrow F_d \xrightarrow{\partial_d} F_{d-1} \longrightarrow \cdots \longrightarrow F_1 \xrightarrow{\partial_1} F_0 \longrightarrow 0$$

be the minimal graded free resolution of M , where $F_i = \bigoplus_j S(-j)^{\beta_{i,j}}$. We set $\nu = \nu_S(M) = \sum_j \beta_{0,j}$ and fix an integer $N > 0$. For the ease of notations in what follows, we will call a column in ∂_i *deep* if all the entries in that column are contained in $\mathfrak{m}^N$ (equivalently, have degree at least N); otherwise, we call the column *exceptional*. This refers to a chosen ordered basis; slightly more formally, a summand $S(-j)$ of F_i is *deep* provided $\partial_i(S(-j)) \subseteq \mathfrak{m}^N F_{i-1}$.

²This is just to avoid some tautology or triviality, and in any case [Theorem 1.3](#) is easy and well-known when $d \leq 1$, see the proof of [Theorem 1.3](#) in [Section 4](#).

Theorem 3.2. *With notations as in [Setting 3.1](#), if there are at most C exceptional columns in the differentials of $F_\bullet$, then*

$$\ell_S(M) \geq (\nu - 3C)N^d.$$

In particular, if all the columns in the differentials of $F_\bullet$ are deep (i.e., $C = 0$), then

$$\ell_S(M) \geq \nu N^d.$$

We first give a quick proof of the “In particular” part of [Theorem 3.2](#) when M is generated in a single degree. We point out that this case is insufficient to deduce the local version even when $C = 0$ (see [Section 4](#)). However, we exhibit the proof as it is an immediate consequence of Boij–Söderberg theory, hinting that the general case might be tractable using ideas and methods behind this theory.

Proof of [Theorem 3.2](#) assuming $C = 0$ and M is generated in a single degree. In this proof, we use $\beta(M)$ to denote the Betti table of M and use $\beta(d^\lambda)$ to denote the pure Betti table corresponding to the degree sequence $d^\lambda = (d_0^\lambda, d_1^\lambda, \dots, d_d^\lambda)$. Without loss of generality we may assume that M is generated in degree 0. By [[ES09](#), Theorem 0.2], we can decompose $\beta(M)$ as a finite sum of various $\beta(d^\lambda)$ with positive rational coefficients, i.e.,

$$\beta(M) = \sum_{\lambda} q_{\lambda} \cdot \beta(d^{\lambda})$$

where $q_{\lambda} \in \mathbb{Q}_{>0}$ and $\sum_{\lambda} q_{\lambda} = \nu$. Now we simply observe that, since all the columns in the differentials of $F_\bullet$ are deep, for all the d^λ 's that show up in the sum above, we must have $d_i^\lambda \geq Ni$. It follows from the multiplicity formula of pure tables [[HM85](#), Theorem 1.2] that

$$\ell_S(M) = \sum_{\lambda} q_{\lambda} \cdot \frac{\prod_{i=1}^d d_i^\lambda}{d!} \geq \sum_{\lambda} q_{\lambda} \cdot \frac{\prod_{i=1}^d (Ni)}{d!} = N^d \sum_{\lambda} q_{\lambda} = \nu N^d. \quad \square$$

In the rest of this section, we shall prove [Theorem 3.2](#), which is significantly more difficult than the special case above. Although the argument itself does not use the Boij–Söderberg decomposition, the technical heart of the proof is similar in spirit to Eisenbud–Schreyer’s positivity argument [[ES09](#), Section 4]. In what follows, we will divide the proof into two parts: in [Section 3.1](#) we state a key cutoff inequality [Lemma 3.4](#) and derive [Theorem 3.2](#) from it; then in [Section 3.2](#) we prove [Lemma 3.4](#).

3.1. The cutoff inequality. We use the notations as in [Setting 3.1](#). We begin by introducing some auxiliary functions. For $0 \leq \tau \leq d - 1$, we define

$$Q_{\tau}(z) = \prod_{\substack{0 \leq h \leq d \\ h \neq \tau, \tau+1}} (z - hN).$$

Note that each $Q_{\tau}(z)$ is a polynomial of degree $d - 1$.

Lemma 3.3. *We have the following identity:*

$$\left(\prod_{h=1}^d (z - hN) \right)'_z = \sum_{\tau=0}^{d-1} \left(\frac{1}{\tau+1} + \dots + \frac{1}{d} \right) Q_{\tau}(z).$$

Proof. This is elementary and follows by induction on $d \geq 2$ and noting that

$$Q_0 + \cdots + Q_{d-1} = d \cdot \prod_{h=1}^{d-1} (z - hN)$$

which itself can also be proved by an easy induction. $\square$

Next, for $t \in \mathbb{R}$, we set

$$(3.3.1) \quad \Phi_\tau(t) = \sum_{i=0}^d (-1)^{i+d-1} \sum_{j \leq t+iN} \beta_{i,j} Q_\tau(j-t).$$

Let $\mathcal{B}_\tau$ be the multiset of degrees of the source summands correspond to exceptional columns in $\partial_\tau: F_\tau \rightarrow F_{\tau-1}$. A summand $S(-b)$ contributes one occurrence of b , and repeated degrees are counted with their multiplicities.

Lemma 3.4 (Cutoff inequality). *For each $0 \leq \tau \leq d-1$ and all $t \in \mathbb{R}$,*

$$\Phi_\tau(t) \geq \sum_{\substack{b \in \mathcal{B}_{\tau+1} \\ \tau N < b-t \leq (\tau+1)N}} -|Q_\tau(b-t)|.$$

In particular, if every column of $\partial_{\tau+1}$ is deep, then $\Phi_\tau(t) \geq 0$ for all t .

In order to deduce [Theorem 3.2](#) from [Lemma 3.4](#), we need to express $\ell_S(M) - \nu N^d$ using certain integration.

Lemma 3.5. *With notations as above, we have*

$$\int_{\mathbb{R}} \left(\sum_{\tau=0}^{d-1} \left(\frac{1}{\tau+1} + \cdots + \frac{1}{d} \right) \Phi_\tau(t) \right) dt = d! (\ell_S(M) - \nu N^d).$$

Proof. By comparing the Hilbert series of $F_\bullet$ and M , we have

$$\sum_{i,j} (-1)^i \beta_{i,j} z^j = (1-z)^d H_M(z).$$

Differentiating and evaluating at $z = 1$, it is easy to see that

$$(3.5.1) \quad \begin{cases} \sum_{i,j} (-1)^i \beta_{i,j} j^h = 0 & \text{if } 0 \leq h < d, \\ \sum_{i,j} (-1)^{i+d} \beta_{i,j} j^d = d! \ell_S(M). \end{cases}$$

From this we can see that each function $\Phi_\tau(t)$ has bounded support: for $t \ll 0$ the sums in [\(3.3.1\)](#) are empty, while for $t \gg 0$ all $\beta_{i,j} Q_\tau(j-t)$ occur in the sum and thus they cancel by [\(3.5.1\)](#), since the coefficients in $Q_\tau(j-t)$ are polynomials in j of degree at most $d-1$.

Choose $T \gg 0$ so that $T \geq j - iN$ for all (i, j) such that $\beta_{i,j} \neq 0$, thus the integrand in the lemma vanishes for $t > T$. It follows that

$$\begin{aligned}
& \int_{\mathbb{R}} \left(\sum_{\tau=0}^{d-1} \left(\frac{1}{\tau+1} + \cdots + \frac{1}{d} \right) \Phi_{\tau}(t) \right) dt \\
&= \sum_{i=0}^d (-1)^{i+d-1} \sum_j \beta_{i,j} \int_{j-iN}^T \left(\sum_{\tau=0}^{d-1} \left(\frac{1}{\tau+1} + \cdots + \frac{1}{d} \right) Q_{\tau}(j-t) \right) dt \\
&= \sum_{i=0}^d (-1)^{i+d-1} \sum_j \beta_{i,j} \left(\prod_{h=1}^d (j-t-hN) \right) \Big|_{t=T}^{t=j-iN} \\
(3.5.2) \quad &= - \sum_j \beta_{0,j} d! N^d - \sum_{i=0}^d (-1)^{i+d-1} \sum_j \beta_{i,j} \prod_{h=1}^d (j-T-hN)
\end{aligned}$$

where the second equality uses [Lemma 3.3](#). Now the first factor in (3.5.2) is $-d! \nu N^d$. Since $\prod_{h=1}^d (j-T-hN)$ is monic of degree d in j , in the second factor of (3.5.2), the terms with j^d sum up to $d! \ell_S(M)$ by (3.5.1) and the lower order terms cancel by (3.5.1). Putting these together, the integral in the lemma equals to

$$d! \ell_S(M) - d! \nu N^d,$$

which is exactly what we want to show. $\square$

Remark 3.6. Note that when there are no exceptional columns, [Lemma 3.4](#) gives $\Phi_{\tau}(t) \geq 0$ for all $0 \leq \tau \leq d-1$, and thus [Lemma 3.5](#) shows that $\ell_S(M) \geq \nu N^d$. This is precisely the “In particular” part of [Theorem 3.2](#).

Proof of Theorem 3.2. By [Lemma 3.5](#), it is enough to show that

$$\int_{\mathbb{R}} \left(\sum_{\tau=0}^{d-1} \left(\frac{1}{\tau+1} + \cdots + \frac{1}{d} \right) \Phi_{\tau}(t) \right) dt \geq -d! \cdot 3CN^d.$$

By [Lemma 3.4](#), each exceptional column in the $(\tau+1)$ -th differential contributes at worst

$$- \int_{\tau N}^{(\tau+1)N} \left(\frac{1}{\tau+1} + \cdots + \frac{1}{d} \right) |Q_{\tau}(u)| du.$$

Since there are at most C exceptional columns among all the differentials, it suffices to show that, for each $0 \leq \tau \leq d-1$,

$$\int_{\tau N}^{(\tau+1)N} \left(\frac{1}{\tau+1} + \cdots + \frac{1}{d} \right) |Q_{\tau}(u)| du \leq d! \cdot 3N^d,$$

which is equivalent to showing that for each $0 \leq \tau \leq d-1$,

$$\frac{1}{d!} \left(\frac{1}{\tau+1} + \cdots + \frac{1}{d} \right) \int_{\tau}^{\tau+1} \prod_{\substack{0 \leq h \leq d \\ h \neq \tau, \tau+1}} |(z-h)| dz \leq 3.$$

This is an elementary calculation: for $\tau \geq 1$ the integrand is bounded by $(\tau + 1)!(d - \tau)!$ and thus the integral is bounded by

$$\frac{(\tau + 1)!(d - \tau)!}{d!} \left(\frac{1}{\tau + 1} + \cdots + \frac{1}{d} \right) \leq 1,$$

while for $\tau = 0$, we have

$$\left(1 + \frac{1}{2} + \cdots + \frac{1}{d}\right) \int_0^1 \prod_{h=2}^d \left(1 - \frac{z}{h}\right) dz \leq \left(1 + \frac{1}{2} + \cdots + \frac{1}{d}\right) \int_0^1 e^{(-\frac{1}{2} - \cdots - \frac{1}{d})z} dz \leq 3. \quad \square$$

3.2. Proof of the cutoff inequality. In this subsection we prove [Lemma 3.4](#). We continue to use the notations as in [Section 3.1](#). Throughout this section, τ will be fixed (as [Lemma 3.4](#) is for any fixed $0 \leq \tau \leq d - 1$).

We shall use cohomology theory of certain supernatural bundles [[ES09](#), Section 6]. Consider the symmetrization map

$$f : (\mathbb{P}_k^1)^{d-1} = \mathbb{P}_k^1 \times \cdots \times \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^{d-1}.$$

This map is linear in the sense that $f^* \mathcal{O}_{\mathbb{P}_k^{d-1}}(1) = \mathcal{O}_{(\mathbb{P}_k^1)^{d-1}}(1, \dots, 1)$. We fix an integer t and consider the following line bundle on $(\mathbb{P}_k^1)^{d-1}$:

$$\mathcal{L} = \boxtimes_{\substack{0 \leq h \leq d \\ h \neq \tau, \tau+1}} \mathcal{O}_{\mathbb{P}_k^1}(t + hN - 1).$$

Note that $\mathcal{L}$ depends on t (we will only use $\mathcal{L}$ when t is a fixed integer in what follows). By the Künneth formula, it is straightforward to see that for every $j \in \mathbb{N}$, $\mathcal{L} \otimes f^* \mathcal{O}(-j)$ has cohomology in at most one degree, namely

$$(3.6.1) \quad q = \#\{h \mid 0 \leq h \leq d, h \neq \tau, \tau + 1, \text{ and } hN < j - t\}.$$

Furthermore, we have

$$(3.6.2) \quad \dim_k H^q((\mathbb{P}_k^1)^{d-1}, \mathcal{L} \otimes f^* \mathcal{O}(-j)) = |Q_\tau(j - t)| = (-1)^{d-1-q} Q_\tau(j - t).$$

Note that, in particular, $\mathcal{L} \otimes f^* \mathcal{O}(-j)$ has no cohomology if $j - t = hN$ for some $0 \leq h \leq d$ and $h \neq \tau, \tau + 1$.

We relate the cohomology of $\mathcal{L} \otimes f^* \mathcal{O}(-j)$ with the function $\Phi_\tau(t)$ that we are interested. Recall that $F_\bullet$ is the minimal graded free resolution of the finite length S -module M . Thus $\tilde{F}_\bullet$ is an exact complex of vector bundles on $\mathbb{P}_k^{d-1}$ and so $f^* \tilde{F}_\bullet \otimes \mathcal{L}$ is an exact complex on $(\mathbb{P}_k^1)^{d-1}$. In particular,

$$R\Gamma((\mathbb{P}_k^1)^{d-1}, f^* \tilde{F}_\bullet \otimes \mathcal{L}) = 0.$$

The E_1 -terms of the spectral sequence computing this derived global section are (henceforth all sheaves are on $(\mathbb{P}_k^1)^{d-1}$ and we omit it in the cohomology) $H^q(f^* \tilde{F}_i \otimes \mathcal{L})$, each of which is a finite dimensional k -vector space.

Lemma 3.7. *For each integer t , set*

$$V = \left(\oplus_{q < \tau} H^q(f^* \tilde{F}_q \otimes \mathcal{L}) \right) \oplus H^\tau(f^* \tilde{F}_\tau \otimes \mathcal{L})^{\leq t + \tau N},$$

$$W = \left(\oplus_{q < \tau} H^q(f^* \tilde{F}_{q+1} \otimes \mathcal{L}) \right) \oplus H^\tau(f^* \tilde{F}_{\tau+1} \otimes \mathcal{L})^{\leq t + (\tau+1)N}$$

where $(-)^{\leq \delta}$ means those summands that come from $\widetilde{S(-j)} \subseteq \widetilde{F}_\tau$ with $j \leq \delta$. Then we have

$$\Phi_\tau(t) = \dim_k V - \dim_k W + \sum_{q-i \leq -2} (-1)^{q-i} \dim_k H^q(f^* \widetilde{F}_i \otimes \mathcal{L}).$$

Proof. Recall from (3.3.1) that

$$\Phi_\tau(t) = \sum_{i=0}^d (-1)^{i+d-1} \sum_{j \leq t+iN} \beta_{i,j} Q_\tau(j-t).$$

By (3.6.2), we have that

$$\sum_q (-1)^{q-i} \dim_k H^q(f^* \widetilde{F}_i \otimes \mathcal{L}) = (-1)^{i+d-1} \sum_j \beta_{i,j} Q_\tau(j-t).$$

We thus count the contribution of $(-1)^{q-i} \dim_k H^q(f^* \widetilde{F}_i \otimes \mathcal{L})$ in computing $\Phi_\tau(t)$. There are four cases to consider.

Case 1: $q-i \leq -2$. In this case by (3.6.1) we have

$$\#\{h \mid 0 \leq h \leq d, h \neq \tau, \tau+1, \text{ and } hN < j-t\} \leq i-2,$$

which implies $j-t \leq iN$. Thus all terms $(-1)^{q-i} \dim_k H^q(f^* \widetilde{F}_i \otimes \mathcal{L})$ appear in $\Phi_\tau(t)$ when $q-i \leq -2$.

Case 2: $q-i > 0$. In this case by (3.6.1) we have

$$\#\{h \mid 0 \leq h \leq d, h \neq \tau, \tau+1, \text{ and } hN < j-t\} > i,$$

which implies $j-t > iN$. Thus none of the terms $(-1)^{q-i} \dim_k H^q(f^* \widetilde{F}_i \otimes \mathcal{L})$ contribute to $\Phi_\tau(t)$ when $q-i > 0$.

Case 3: $q-i = 0$. In this case, if $q = i < \tau$, then from

$$\#\{h \mid 0 \leq h \leq d, h \neq \tau, \tau+1, \text{ and } hN < j-t\} = i$$

we can conclude that $j-t \leq iN$. Thus all terms $\dim_k H^q(f^* \widetilde{F}_q \otimes \mathcal{L})$ appear in $\Phi_\tau(t)$ when $q < \tau$. Otherwise if $q = i \geq \tau$, then we must have $q = i = \tau$ in order to obtain nontrivial contribution to $\Phi_\tau(t)$ (by analyzing the number of h 's), in which case the contribution to $\Phi_\tau(t)$ is precisely

$$\dim_k H^\tau(f^* \widetilde{F}_\tau \otimes \mathcal{L})^{\leq t+\tau N} = (-1)^{\tau+d-1} \sum_{j \leq t+\tau N} \beta_{\tau,j} Q_\tau(j-t).$$

Case 4: $q-i = -1$. This case is similar to the case above so we omit details: if $q = i-1 < \tau$, then all terms $\dim_k H^q(f^* \widetilde{F}_{q+1} \otimes \mathcal{L})$ appear in $\Phi_\tau(t)$, while if $q = i-1 = \tau$ (which is the only other possibility of obtaining nontrivial contribution), then the contribution to $\Phi_\tau(t)$ is precisely

$$-\dim_k H^\tau(f^* \widetilde{F}_{\tau+1} \otimes \mathcal{L})^{\leq t+(\tau+1)N} = (-1)^{\tau+d} \sum_{j \leq t+(\tau+1)N} \beta_{\tau+1,j} Q_\tau(j-t).$$

Putting these four cases together proves the lemma (V and W correspond to the contributions in Case 3 and Case 4 respectively). $\square$

We can now use Lemma 3.7 to establish the cutoff inequality Lemma 3.4.

Proof of Lemma 3.4. We first consider the case that t is an integer. By Lemma 3.7, it suffices to show that

$$(3.7.1) \quad \dim_k V - \dim_k W + \sum_{q-i \leq -2} (-1)^{q-i} \dim_k H^q(f^* \tilde{F}_i \otimes \mathcal{L}) \geq \sum_{\substack{b \in \mathcal{B}_{\tau+1} \\ \tau N < b-t \leq (\tau+1)N}} -|Q_\tau(b-t)|.$$

We consider the spectral sequence

$$E_1^{p,q} = H^q(f^* \tilde{F}_{-p} \otimes \mathcal{L}) \implies H^{p+q} R\Gamma(f^* \tilde{F}_\bullet \otimes \mathcal{L})$$

and set $U^n = \bigoplus_{p+q=n} E_1^{p,q}$ for each n . Since this spectral sequence arises from a bicomplex of vector spaces, we may apply [EFS03, Lemma 3.5] to obtain a degree one map $d : U^\bullet \rightarrow U^{\bullet+1}$ with the following properties:

- $(U^\bullet, d)$ is a complex of k -vector spaces whose homology coincides with the abutment of the spectral sequence;
- for all p, q we have $d(E_1^{p,q}) \subseteq \bigoplus_{r \geq 1} E_1^{p+r, q-r+1}$;
- the composition of $E_1^{p,q} \rightarrow U^{p+q+1} \twoheadrightarrow E_1^{p+1, q}$ is the differential in the E_1 -page of the spectral sequence.

In our situation, since $R\Gamma(f^* \tilde{F}_\bullet \otimes \mathcal{L}) = 0$, the complex $(U^\bullet, d)$ is a bounded, exact complex of finite dimensional vector spaces. We claim that for any subspaces $U_\circ^{-1} \subseteq U^{-1}$ and $U_\circ^0 \subseteq U^0$ we have

$$\dim_k U_\circ^0 - \dim_k U_\circ^{-1} + \sum_{j \leq -2} (-1)^j \dim_k U_j \geq -\dim_k \left(\frac{U_\circ^{-1}}{U_\circ^{-1} \cap d^{-1}(U_\circ^0)} \right),$$

where d^{-1} refers to taking the inverse image of subspaces under the map $d : U^{-1} \rightarrow U^0$ (not the degree -1 component of the differential). This holds since

$$\dots \rightarrow U^{-3} \rightarrow U^{-2} \rightarrow d^{-1}(U_\circ^0) \rightarrow U_\circ^0$$

is exact, so that $\dim_k U_\circ^0 - \dim_k d^{-1}(U_\circ^0) + \sum_{j \leq -2} (-1)^j \dim_k U_j \geq 0$, together with the fact that $\dim_k d^{-1}(U_\circ^0) - \dim_k U_\circ^{-1} \geq -\dim_k \left(\frac{U_\circ^{-1}}{U_\circ^{-1} \cap d^{-1}(U_\circ^0)} \right)$.

We apply this with $U_\circ^0 = V$ and $U_\circ^{-1} = W$ to obtain

$$(3.7.2) \quad \dim_k V - \dim_k W + \sum_{q-i \leq -2} (-1)^{q-i} \dim_k H^q(f^* \tilde{F}_i \otimes \mathcal{L}) \geq -\dim_k \left(\frac{W}{W \cap d^{-1}(V)} \right).$$

From the description of the differential d given above and the definitions of V and W , it is easy to see that

$$\frac{W}{W \cap d^{-1}(V)} = \frac{H^\tau(f^* \tilde{F}_{\tau+1} \otimes \mathcal{L})^{\leq t+(\tau+1)N}}{H^\tau(f^* \tilde{F}_{\tau+1} \otimes \mathcal{L})^{\leq t+(\tau+1)N} \cap W'}$$

where W' is the inverse image of $H^\tau(f^* \tilde{F}_\tau \otimes \mathcal{L})^{\leq t+\tau N}$ under the E^1 differential $H^\tau(f^* \tilde{F}_{\tau+1} \otimes \mathcal{L}) \rightarrow H^\tau(f^* \tilde{F}_\tau \otimes \mathcal{L})$. It follows that $\frac{W}{W \cap d^{-1}(V)}$ is a quotient of

$$\bigoplus_{\substack{b \in \mathcal{B}_{\tau+1}, \\ t+\tau N < b \leq t+(\tau+1)N}} H^\tau(f^* \mathcal{O}(-b) \otimes \mathcal{L}).$$

Thus (3.7.1) follows from (3.7.2) and (3.6.2).

This completes the proof of [Lemma 3.4](#) when t is an integer. To prove [Lemma 3.4](#) for general real number t , we observe that both functions in this inequality are right continuous and thus we may assume t is rational. Choose an integer $L \gg 0$ so that tL is an integer. We consider the faithfully flat base change (the k -linear L -th “Frobenius”):

$$\sigma_L : S \rightarrow S \quad x_i \mapsto x_i^L.$$

Then $\sigma_L^* F_\bullet$ is the minimal graded free resolution of $\sigma_L^* M$. It has the same Betti numbers with all internal degrees j multiplied by L , and the deep (resp., exceptional) columns remain deep (resp., exceptional) with respect to $\mathfrak{m}^{NL}$. We now apply the already established [Lemma 3.4](#) to $\sigma_L^* M$, $\sigma_L^* F_\bullet$, tL and NL . It is straightforward to check that both sides of the inequality in [Lemma 3.4](#) are simply scaled by a factor of L^{d-1} . This completes the proof of [Lemma 3.4](#). $\square$

4. PROOF OF [THEOREM 1.3](#): THE GENERAL CASE

In this section we complete the proof of [Theorem 1.3](#). The technical heart in this section is [Theorem 4.2](#), which essentially allows us to reduce to the graded case.

Setting 4.1. Let $S = k[x_1, \dots, x_d]$ be a polynomial ring over a field k and $\mathfrak{m} = (x_1, \dots, x_d)$. We assume $d \geq 2$. Let M be a finitely generated S -module supported only at $\{\mathfrak{m}\}$, and let $F_\bullet$ be a free resolution of M over S such that $\partial(F_\bullet) \subseteq \mathfrak{m}F_\bullet$. Fix an integer $N > 0$. For a chosen basis of each F_i (which will be fixed), we call a column in a differential of $F_\bullet$ deep if all entries in that column are contained in $\mathfrak{m}^N$, otherwise, we call the column exceptional.

Theorem 4.2. *With notations as in [Setting 4.1](#), if there are at most C exceptional columns in the differentials of $F_\bullet$. Then for every integer $L > 0$, there exists a graded S -module M_L of finite length with a minimal graded free resolution $F_{L,\bullet}$ (and a chosen basis for each $F_{L,i}$) such that when $L \rightarrow \infty$, we have*

$$\begin{aligned} \nu_S(M_L) &= L^d \nu_S(M), \\ \ell_S(M_L) &= L^d \ell_S(M) + O(L^{d-1}), \\ C_L &\leq L^d C + O(L^{d-1}) \end{aligned}$$

where C_L is the total number of exceptional columns in the differentials of $F_{L,\bullet}$.

Proof. We first construct M_L . For ease of notations, in the proof we will denote the original polynomial ring as $S_0 = k[x_1, \dots, x_d]$ (M and $F_\bullet$ are thus defined over S_0). Consider the ring homomorphism

$$S_0 \rightarrow T := k[y_1, \dots, y_d, z_1, \dots, z_d] \quad x_i \mapsto y_i z_i.$$

Let $M'_L := M \otimes_{S_0} T / (z_1^L, \dots, z_d^L)$. Fix an integer $K > 0$ so that $\mathfrak{m}_{S_0}^K M = 0$. Set

$$M_L := M'_L / (y_1, \dots, y_d)^K M'_L = M \otimes_{S_0} T / ((z_1^L, \dots, z_d^L) + (y_1, \dots, y_d)^K).$$

We will view M_L as a module over $S := k[y_1, \dots, y_d]$ (and $\mathfrak{m}_S := (y_1, \dots, y_d)$ in what follows).

We need to show that M_L satisfies the desired properties. We first note that M_L is graded (in fact, it is $\mathbb{Z}^d$ -graded). To see this, note that we can assign a $\mathbb{Z}^d$ -grading on T :

$$\deg(y_i) = e_i, \quad \deg(z_i) = -e_i, \quad 1 \leq i \leq d$$

where e_i is the i -th standard basis of $\mathbb{Z}^d$. With this grading, each $y_i z_i$ has degree 0 and thus $M \otimes_{S_0} T$ is $\mathbb{Z}^d$ -graded: $M = \text{coker}(F_1 \xrightarrow{\partial_1} F_0)$ where the matrix representing ∂_1 has entries in $\mathfrak{m}_{S_0}$ and thus after base change to T , all entries in the presentation matrix of $M \otimes_{S_0} T$

are polynomials in $y_i z_i$ so they are multi-homogeneous (in fact they all have multi-degree 0) and we can assign the chosen basis of F_1 and F_0 multi-degree 0 too. It follows that M'_L and M_L are also $\mathbb{Z}^d$ -graded, since they are $\mathbb{Z}^d$ -graded quotient of $M \otimes_{S_0} T$. Clearly, M_L is only supported at $\{\mathfrak{m}_S\}$ and thus has finite length over S (we caution the readers that M'_L may not have finite length over S).

We next observe that $\nu_S(M_L) = L^d \nu_{S_0}(M)$. This is because we have an exact sequence:

$$F_1 \otimes_{S_0} T/(z_1^L, \dots, z_d^L) \xrightarrow{\partial_1} F_0 \otimes_{S_0} T/(z_1^L, \dots, z_d^L) \rightarrow M'_L \rightarrow 0.$$

Since all the entries in ∂_1 are polynomials in $y_i z_i$, $\text{Image}(\partial_1) \subseteq \mathfrak{m}_S(F_0 \otimes_{S_0} T/(z_1^L, \dots, z_d^L))$ and thus

$$\nu_S(M_L) = \nu_S(M'_L) = \nu_S(F_0 \otimes_{S_0} T/(z_1^L, \dots, z_d^L)) = L^d \text{rank}_{S_0}(F_0) = L^d \nu_{S_0}(M).$$

Next we will show that $\ell_S(M_L) = L^d \ell_{S_0}(M) + O(L^{d-1})$. We will introduce another auxiliary module $M''_L := M \otimes_{S_0} T/(z_1^L - 1, \dots, z_d^L - 1)$ and compare $\ell_S(M_L)$ with $\ell_S(M''_L)$. Note that M''_L is already annihilated by $\mathfrak{m}_S^K$: this is because M''_L is annihilated by $\mathfrak{m}_{S_0}^K T = (y_1 z_1, \dots, y_d z_d)^K$ by base change, but as $z_1, \dots, z_d$ are units in $T/(z_1^L - 1, \dots, z_d^L - 1)$,

$$\mathfrak{m}_S^K M''_L = (y_1, \dots, y_d)^K M''_L = (y_1 z_1, \dots, y_d z_d)^K M''_L = 0.$$

We now compare the presentations of M_L and M''_L over $S/\mathfrak{m}_S^K$:

$$F_1 \otimes_{S_0} T/((z_1^L, \dots, z_d^L) + \mathfrak{m}_S^K T) \xrightarrow{\partial'_1} F_0 \otimes_{S_0} T/((z_1^L, \dots, z_d^L) + \mathfrak{m}_S^K T) \rightarrow M_L \rightarrow 0.$$

$$F_1 \otimes_{S_0} T/((z_1^L - 1, \dots, z_d^L - 1) + \mathfrak{m}_S^K T) \xrightarrow{\partial''_1} F_0 \otimes_{S_0} T/((z_1^L - 1, \dots, z_d^L - 1) + \mathfrak{m}_S^K T) \rightarrow M''_L \rightarrow 0$$

where ∂'_1 and ∂''_1 are simply the base changes of $F_1 \xrightarrow{\partial_1} F_0$.

Let $\{f_{0,j}\}_{j \in J_0}$ and $\{f_{1,j}\}_{j \in J_1}$ be the chosen free basis of F_0 and F_1 (over S_0) respectively. Then in both presentations above, the source and target are free modules over $S/\mathfrak{m}_S^K$ with basis $\{\underline{z}^{\underline{u}} f_{1,j}\}_{j \in J_1, \underline{u} < \underline{L}}$ and $\{\underline{z}^{\underline{u}} f_{0,j}\}_{j \in J_0, \underline{u} < \underline{L}}$ respectively, where $\underline{z}^{\underline{u}}$ denotes $z_1^{u_1} \cdots z_d^{u_d}$ and $\underline{u} < \underline{L}$ means $u_i < L$ for all i . Let $E > 0$ be an integer that bounds the degrees of all monomials (in x_i 's) that show up in the original differential $F_1 \xrightarrow{\partial_1} F_0$. We examine the following diagram

$$\begin{array}{ccc} \bigoplus_{j \in J_1, \underline{u} < \underline{L}} (S/\mathfrak{m}_S^K) \cdot \underline{z}^{\underline{u}} f_{1,j} & \begin{array}{c} \xrightarrow{\partial'_1} \\ \xrightarrow{\partial''_1} \end{array} & \bigoplus_{j \in J_0, \underline{u} < \underline{L}} (S/\mathfrak{m}_S^K) \cdot \underline{z}^{\underline{u}} f_{0,j} \\ \uparrow & \nearrow \partial'_1 = \partial''_1 & \\ \bigoplus_{j \in J_1, \underline{u} < \underline{L}-E} (S/\mathfrak{m}_S^K) \cdot \underline{z}^{\underline{u}} f_{1,j} & & \end{array}$$

The key observation here is that the two maps ∂'_1 and ∂''_1 agree on the free $S/\mathfrak{m}_S^K$ -submodule with basis $\{\underline{z}^{\underline{u}} f_{1,j}\}_{j \in J_1, \underline{u} < \underline{L}-E}$, as indicated by the dashed arrow above. To see this, note that either map sends $\underline{z}^{\underline{u}} f_{1,j}$ to a k -linear combination of

$$\underline{y}^{\underline{v}} \underline{z}^{\underline{v}+\underline{u}} f_{0,j} \in \bigoplus_{j \in J_0, \underline{u} < \underline{L}} (S/\mathfrak{m}_S^K) \cdot \underline{z}^{\underline{u}} f_{0,j}$$

where for ∂'_1 , $\underline{z}^{\underline{v}+\underline{u}} = 0$ if $\underline{v} + \underline{u} \not< \underline{L}$ and for ∂''_1 , $\underline{z}^{\underline{v}+\underline{u}} = \underline{z}^{\underline{w}}$ where $\underline{w} = \underline{v} + \underline{u} \bmod \underline{L}$. But by our choice of E , all possible $\underline{v}$'s that occur in the linear combination satisfy $\underline{v} \leq \underline{E}$. It follows

that for $\underline{u} < \underline{L} - E$, the images of $\underline{z}^{\underline{u}} f_{1,j}$ under ∂'_1 and ∂''_1 are the same (the distinction above does not occur as $\underline{v} + \underline{u} < \underline{L}$). Thus, by the diagram above, we have

$$\begin{aligned} |\ell_S(M_L) - \ell_S(M''_L)| &= |\ell_S(\text{coker } \partial'_1) - \ell_S(\text{coker } \partial''_1)| \\ &\leq \ell_S \left(\bigoplus_{j \in J_1, \underline{u} < \underline{L}} (S/\mathfrak{m}_S^K) \cdot \underline{z}^{\underline{u}} f_{1,j} \right) - \ell_S \left(\bigoplus_{j \in J_1, \underline{u} < \underline{L} - E} (S/\mathfrak{m}_S^K) \cdot \underline{z}^{\underline{u}} f_{1,j} \right) \\ &= \text{rank}_{S_0}(F_1) \ell_S(S/\mathfrak{m}_S^K) \cdot (L^d - (L - E)^d) \\ &= O(L^{d-1}) \end{aligned}$$

since $F_\bullet$ (and in particular $\text{rank}_{S_0}(F_1)$), K , and E do not depend on L . It remains to observe that $\ell_S(M''_L) = L^d \ell_{S_0}(M)$, which follows as the extension

$$S_0 \rightarrow T/(z_1^L - 1, \dots, z_d^L - 1) \quad x_i \mapsto y_i z_i$$

is finite free of rank L^d .

Finally, we shall construct a minimal graded free resolution $F_{L,\bullet}$ of M_L with at most $L^d C + O(L^{d-1})$ exceptional columns in its differentials (for a chosen basis for each $F_{L,i}$). Consider the following short exact sequence:

$$0 \rightarrow \mathfrak{m}_S^K(M \otimes_{S_0} T) \rightarrow M \otimes_{S_0} T \rightarrow (M \otimes_{S_0} T)/\mathfrak{m}_S^k(M \otimes_{S_0} T) \rightarrow 0.$$

After derived tensor with $T/(z_1^L, \dots, z_d^L)$, we obtain an exact triangle:

(4.2.1)

$$\mathfrak{m}_S^K(M \otimes_{S_0} T)/^{\mathbb{L}}(z_1^L, \dots, z_d^L) \xrightarrow{\alpha} (M \otimes_{S_0} T)/^{\mathbb{L}}(z_1^L, \dots, z_d^L) \rightarrow \frac{(M \otimes_{S_0} T)}{\mathfrak{m}_S^k(M \otimes_{S_0} T)}/^{\mathbb{L}}(z_1^L, \dots, z_d^L) \xrightarrow{+1}.$$

Note that by construction M_L is the 0-th homology of the last term above. We will analyze the total Betti numbers (over S) of $\mathfrak{m}_S^K(M \otimes_{S_0} T)/^{\mathbb{L}}(z_1^L, \dots, z_d^L)$ as well as the fiber of

$$\frac{(M \otimes_{S_0} T)}{\mathfrak{m}_S^k(M \otimes_{S_0} T)}/^{\mathbb{L}}(z_1^L, \dots, z_d^L) \rightarrow M_L.$$

Note that $z_1^K \cdots z_d^K$ annihilates $\mathfrak{m}_S^K(M \otimes_{S_0} T)$: this is because $(z_1^K \cdots z_d^K) \mathfrak{m}_S^K T \subseteq \mathfrak{m}_{S_0}^K T$ and $\mathfrak{m}_{S_0}^K$ annihilates M by our choice of K . To estimate the total Betti numbers over S , we observe that

$$\begin{aligned} (\mathfrak{m}_S^K(M \otimes_{S_0} T)/^{\mathbb{L}}(z_1^L, \dots, z_d^L)) \otimes_S^{\mathbb{L}} S/\mathfrak{m}_S &\cong (\mathfrak{m}_S^K(M \otimes_{S_0} T)/^{\mathbb{L}}(z_1^L, \dots, z_d^L)) \otimes_T^{\mathbb{L}} T/\mathfrak{m}_S T \\ (4.2.2) \quad &\cong \mathfrak{m}_S^K(M \otimes_{S_0} T) \otimes_T^{\mathbb{L}} T/(y_1, \dots, y_d, z_1^L, \dots, z_d^L) \end{aligned}$$

Consider a minimal graded free resolution of $\mathfrak{m}_S^K(M \otimes_{S_0} T)$ over T with i -th Betti number b_i , the i -th homology of (4.2.2) is the homology of

$$(T/(y_1, \dots, y_d, z_1^L, \dots, z_d^L))^{b_{i+1}} \xrightarrow{\gamma} (T/(y_1, \dots, y_d, z_1^L, \dots, z_d^L))^{b_i} \rightarrow (T/(y_1, \dots, y_d, z_1^L, \dots, z_d^L))^{b_{i-1}}.$$

Since we also know that this homology is annihilated by $z_1^K \cdots z_d^K$, its dimension over $S/\mathfrak{m}_S$ (equivalently, its length over T) is bounded by

$$\ell_T \left(\frac{\text{Image } \gamma : (z_1^K \cdots z_d^K)}{\text{Image } \gamma} \right).$$

By considering the exact sequence

$$0 \rightarrow \frac{\text{Image } \gamma : (z_1^K \cdots z_d^K)}{\text{Image } \gamma} \rightarrow \frac{(T/(y_1, \dots, y_d, z_1^L, \dots, z_d^L))^{b_i}}{\text{Image } \gamma} \xrightarrow{\cdot(z_1^K \cdots z_d^K)} \frac{(T/(y_1, \dots, y_d, z_1^L, \dots, z_d^L))^{b_i}}{\text{Image } \gamma} \rightarrow \text{coker} \rightarrow 0,$$

we find that

$$\ell_T \left(\frac{\text{Image } \gamma : (z_1^K \cdots z_d^K)}{\text{Image } \gamma} \right) = \ell_T(\text{coker}) \leq b_i \cdot \ell_T \left(\frac{T}{(y_1, \dots, y_d, z_1^L, \dots, z_d^L, z_1^K \cdots z_d^K)} \right) = O(L^{d-1})$$

where the last equality follows as K and b_i do not depend on L . Since this is true for each i , we find that the sum of the dimensions of all the homologies of (4.2.2) (note that there are at most $2d$ nonzero homologies) is $O(L^{d-1})$, which proves that

$$(4.2.3) \quad \beta_S(\mathfrak{m}_S^K(M \otimes_{S_0} T)/^{\mathbb{L}}(z_1^L, \dots, z_d^L)) = O(L^{d-1}).$$

We next consider the exact triangle defining W :

$$(4.2.4) \quad W \rightarrow \frac{(M \otimes_{S_0} T)}{\mathfrak{m}_S^K(M \otimes_{S_0} T)}/^{\mathbb{L}}(z_1^L, \dots, z_d^L) \rightarrow M_L \xrightarrow{+1}$$

and we claim that the total Betti numbers of W over S is $O(L^{d-1})$. To prove this, we rewrite the derived tensor as

$$\frac{(M \otimes_{S_0} T)}{\mathfrak{m}_S^K(M \otimes_{S_0} T)}/^{\mathbb{L}}(z_1^L, \dots, z_d^L) \cong \frac{(M \otimes_{S_0} T)}{\mathfrak{m}_S^K(M \otimes_{S_0} T)} \otimes_{k[z_1, \dots, z_d]}^{\mathbb{L}} \frac{k[z_1, \dots, z_d]}{(z_1^L, \dots, z_d^L)}.$$

Note that the module $(M \otimes_{S_0} T)/\mathfrak{m}_S^K(M \otimes_{S_0} T)$ is finitely generated graded over $k[z_1, \dots, z_d]$, thus we can find a free $k[z_1, \dots, z_d]$ -module together with an exact triangle

$$k[z_1, \dots, z_d]^{\oplus n} \rightarrow (M \otimes_{S_0} T)/\mathfrak{m}_S^K(M \otimes_{S_0} T) \rightarrow Q \xrightarrow{+1}$$

such that Q is annihilated by some (homogeneous) $0 \neq g \in k[z_1, \dots, z_d]$ in $D(k[z_1, \dots, z_d])$. It follows that for all $i \geq 1$,

$$H_i \left(\frac{(M \otimes_{S_0} T)}{\mathfrak{m}_S^K(M \otimes_{S_0} T)} \otimes_{k[z_1, \dots, z_d]}^{\mathbb{L}} \frac{k[z_1, \dots, z_d]}{(z_1^L, \dots, z_d^L)} \right)$$

is annihilated by g . Consider a minimal graded free resolution of $(M \otimes_{S_0} T)/\mathfrak{m}_S^K(M \otimes_{S_0} T)$ over $k[z_1, \dots, z_d]$, with i -th Betti number c_i , the i -th homology above is

$$\left(\frac{k[z_1, \dots, z_d]}{(z_1^L, \dots, z_d^L)} \right)^{c_{i+1}} \rightarrow \left(\frac{k[z_1, \dots, z_d]}{(z_1^L, \dots, z_d^L)} \right)^{c_i} \rightarrow \left(\frac{k[z_1, \dots, z_d]}{(z_1^L, \dots, z_d^L)} \right)^{c_{i-1}}.$$

By the same analysis as above (when we deal with (4.2.2)), the k -vector space dimension of this homology is bounded by $c_i \cdot \dim_k(k[z_1, \dots, z_d]/(z_1^L, \dots, z_d^L, g)) = O(L^{d-1})$, since c_i and g do not depend on L . It follows that for all $i \geq 1$,

$$\ell_T \left(H_i \left(\frac{(M \otimes_{S_0} T)}{\mathfrak{m}_S^K(M \otimes_{S_0} T)}/^{\mathbb{L}}(z_1^L, \dots, z_d^L) \right) \right) = O(L^{d-1}).$$

Therefore we have

$$(4.2.5) \quad \beta_S(W) \leq \beta_S(k) \cdot \sum_i \ell_S(H_i(W))$$

$$= \beta_S(k) \cdot \sum_{i \geq 1} \ell_T \left(H_i \left(\frac{(M \otimes_{S_0} T)}{\mathfrak{m}_S^K(M \otimes_{S_0} T)} /^{\mathbb{L}}(z_1^L, \dots, z_d^L) \right) \right) = O(L^{d-1}).$$

We now put everything together. The middle term of (4.2.1) can be identified with $F_\bullet \otimes_{S_0} T/(z_1^L, \dots, z_d^L)$ and by our assumption, there are at most C exceptional columns in the differentials of $F_\bullet$ and from this we know that there are at most $L^d C$ exceptional columns in the differentials of $F_\bullet \otimes_{S_0} T/(z_1^L, \dots, z_d^L)$, viewed as a graded finite free complex over S and with the basis the original chosen basis of F_i times $\underline{z}^{\underline{u}}$ for $\underline{u} < \underline{L}$ (the entries in the differential of this complex are the monomials that show up in the original differential of $F_\bullet$). By (4.2.3), the leftmost term in (4.2.1) has total Betti numbers at most $O(L^{d-1})$, therefore by replacing the leftmost term by its minimal graded free resolution over S , the mapping cone of α , which is isomorphic to the rightmost term of (4.2.1), is a graded free resolution of the latter over S with at most $L^d C + O(L^{d-1})$ exceptional columns (this follows by examining the differentials in the mapping cone). Finally, we examine the exact triangle (4.2.4), we have just shown that the middle term admits a graded free resolution over S with at most $L^d C + O(L^{d-1})$ exceptional columns under a chosen basis (so we replace the middle term by this complex). But by (4.2.5), the leftmost term in (4.2.4) has total Betti numbers at most $O(L^{d-1})$. Thus by the same argument as above, i.e., replacing the leftmost term by its minimal graded free resolution over S , we find that the mapping cone is a graded free resolution of M_L with at most $L^d C + O(L^{d-1})$ exceptional columns under appropriate chosen basis. Trimming this mapping cone to a minimal graded free resolution $F_{L,\bullet}$ then completes the proof of the theorem. $\square$

Finally, we prove Theorem 1.3 in the general case, using Theorem 4.2 to reduce it to the graded case Theorem 3.2.

Proof of Theorem 1.3. If $d = 0$ the conclusion is trivial. If $d = 1$, then M is the cokernel of $A^n \xrightarrow{X} A^n$ (we may assume that $n = \nu_A(M)$) and it is well-known that

$$\ell_A(M) = \ell_A(A/\det(X)) \geq \ell_A(A/\mathfrak{m}_A^{(n-C)N}) = (\nu_A(M) - C)N \geq (\nu_A(M) - 3C)N.$$

Henceforth we assume that $d \geq 2$. We may assume A is complete and thus by Cohen's structure theorem, we have $A \cong k[[x_1, \dots, x_d]]$ where k is a coefficient field of A . Let $F_\bullet$ be the free resolution of M over A that satisfies the properties in the statement of the theorem (i.e., there are at most C exceptional columns in the differentials of $F_\bullet$ under a chosen basis of each F_i). By trimming $F_\bullet$ we may assume that $F_\bullet$ is minimal.

We next claim that $M^{\oplus d}$, when viewed as a finite length module over $S = k[x_1, \dots, x_d] \subseteq A$ supported only at $\mathfrak{m} = (x_1, \dots, x_d)$, admits a finite free resolution $G_\bullet$ over S so that all the entries in the differentials are contained in $\mathfrak{m}$. In particular, $G_\bullet \otimes_S A$ will be a minimal free resolution of $M^{\oplus d}$ over A . Let n_i be the i -th Betti number of M over A . We construct $G_\bullet$ step by step, starting with a surjection $S^{dn_0} \twoheadrightarrow M^{\oplus d}$. At the i -th stage, suppose

$$S^{dn_{i-1}} \xrightarrow{\partial_{i-1}} S^{dn_{i-2}} \rightarrow \dots \rightarrow S^{dn_0} \rightarrow M^{\oplus d} \rightarrow 0$$

has been constructed. We consider the i -th syzygy module $\text{Syz}_S^i(M) = \ker(\partial_{i-1})$. By the minimality already established, we know that $\widehat{\text{Syz}_S^i(M^{\oplus d})} \cong \text{Syz}_A^i(M^{\oplus d})$ and thus

$$(4.2.6) \quad \text{rank}_S \text{Syz}_S^i(M^{\oplus d}) = \text{rank}_A \text{Syz}_A^i(M^{\oplus d}) = dn_i - \text{rank}_A \text{Syz}_A^{i+1}(M^{\oplus d}) \leq dn_i - d$$

when $i \leq d-1$. By [Swa67], we know that

$$\nu_S(\text{Syz}_S^i(M^{\oplus d})) \leq \sup_P \{ \nu_{S_P}(\text{Syz}_S^i(M^{\oplus d})_P) + \dim(S/P) \}.$$

We examine the right hand side of the above: for $P = \mathfrak{m}$, it gives dn_i (as this is the minimal number of generators of its completion $\text{Syz}_A^i(M^{\oplus d})$); while for $P \neq \mathfrak{m}$, $\text{Syz}_S^i(M^{\oplus d})_P$ is free and (4.2.6) shows that its rank is $\leq dn_i - d$. It follows that $\nu_S(\text{Syz}_S^i(M^{\oplus d})) = dn_i$, i.e., we can find a surjection $S^{dn_i} \twoheadrightarrow \text{Syz}_S^i(M^{\oplus d})$ and thus continue this procedure (the entries in the differential $S^{dn_i} \xrightarrow{\partial_i} S^{dn_{i-1}}$ are guaranteed to be in $\mathfrak{m}$ as the Betti numbers are the same as the Betti numbers in the minimal free resolution over A). Finally, at the d -th stage, $\text{Syz}_S^d(M^{\oplus d})$ is locally free and thus free over S by the Quillen–Suslin theorem [Qui76, Sus76]. But then it is isomorphic to S^{dn_d} by comparing its rank with the Betti numbers over A .

We can now replace M by $M^{\oplus d}$ and $F_\bullet$ by $F_\bullet^{\oplus d}$ (under such replacements, the length, the minimal number of generators, as well as C are all scaled by d which does not affect the conclusion) to assume that M admits a finite free resolution $G_\bullet$ over S such that $\partial(G_\bullet) \subseteq \mathfrak{m}G_\bullet$. Note that $G_\bullet \otimes_S A$ and $F_\bullet$ are both minimal free resolutions of M over A . Thus, if X_i and Y_i represent the matrices in the i -th differential of $G_\bullet$ and $F_\bullet$ respectively (under a chosen basis), then there exists $U_i \in \text{SL}_{n_i}(A)$ such that $U_i X_i U_i^{-1} = Y_i$. Choose $V_i \in \text{SL}_{n_i}(S)$ such that $V_i \equiv U_i \pmod{\mathfrak{m}^N}$ (this is possible since $\text{SL}_n(S) \rightarrow \text{SL}_n(S/\mathfrak{m}^N)$ is surjective). We then have $V_i X_i V_i^{-1} \equiv Y_i \pmod{\mathfrak{m}^N}$. Therefore, after we change basis for each G_i via V_i , we may assume that there are at most C exceptional columns in the differentials of $G_\bullet$ (as the same holds for $F_\bullet$ with the chosen basis by assumption). We are now in a situation to apply Theorem 4.2, which gives us graded S -modules M_L of finite length together with their minimal graded free resolution $F_{L,\bullet}$ so that

$$\nu_S(M_L) = L^d \nu_S(M),$$

$$\ell_S(M_L) = L^d \ell_S(M) + O(L^{d-1}),$$

$$C_L \leq L^d C + O(L^{d-1}).$$

Applying Theorem 3.2 to each M_L , we obtain that

$$\ell_S(M_L) \geq (\nu_S(M_L) - 3C_L)N^d.$$

Putting these together we find that

$$L^d \ell_S(M) \geq L^d \nu_S(M)N^d - 3L^d C N^d - O(L^{d-1}).$$

Divide by L^d and let $L \rightarrow \infty$, we obtain that

$$\ell_S(M) \geq \nu_S(M)N^d - 3C N^d.$$

Since the length and minimal number of generators computed over S and A are clearly the same, this completes the proof of the theorem. $\square$

REFERENCES

- [BHM26] B. BHATT, M. HOCHSTER, AND L. MA: *Lim Cohen-Macaulay sequences of modules*, J. Reine Angew. Math. **835** (2026), 111–169. 5080046
- [EFS03] D. EISENBUD, G. FLØYSTAD, AND F.-O. SCHREYER: *Sheaf cohomology and free resolutions over exterior algebras*, Transactions of the American Mathematical Society **355** (2003), no. 11, 4397–4426.
- [ES09] D. EISENBUD AND F.-O. SCHREYER: *Betti numbers of graded modules and cohomology of vector bundles*, J. Amer. Math. Soc. **22** (2009), no. 3, 859–888. 2505303
- [HM85] C. HUNEKE AND M. MILLER: *A note on the multiplicity of Cohen-Macaulay algebras with pure resolutions*, Canad. J. Math. **37** (1985), no. 6, 1149–1162. 828839
- [IMW22] S. B. IYENGAR, L. MA, AND M. E. WALKER: *Multiplicities and Betti numbers in local algebra via lim Ulrich points*, Algebra Number Theory **16** (2022), no. 5, 1213–1257. 4471041
- [Lec60] C. LECH: *Note on multiplicities of ideals*, Ark. Mat. **4** (1960), 63–86 (1960). 140536
- [Ma17] L. MA: *Lech’s conjecture in dimension three*, Adv. Math. **322** (2017), 940–970. 3720812
- [PS73] C. PESKINE AND L. SZPIRO: *Dimension projective finie et cohomologie locale. Applications à la démonstration de conjectures de M. Auslander, H. Bass et A. Grothendieck*, Inst. Hautes Études Sci. Publ. Math. (1973), no. 42, 47–119. 374130
- [Qui76] D. QUILLEN: *Projective modules over polynomial rings*, Invent. Math. **36** (1976), 167–171. 427303
- [Rob89] P. ROBERTS: *Intersection theorems*, Commutative algebra (Berkeley, CA, 1987), Math. Sci. Res. Inst. Publ., vol. 15, Springer, New York, 1989, pp. 417–436. 1015532
- [Rob98] P. C. ROBERTS: *Multiplicities and Chern classes in local algebra*, Cambridge Tracts in Mathematics, vol. 133, Cambridge University Press, Cambridge, 1998. 1686450
- [Sei89] G. SEIBERT: *Complexes with homology of finite length and Frobenius functors*, J. Algebra **125** (1989), no. 2, 278–287. 1018945
- [Sus76] A. A. SUSLIN: *Projective modules over polynomial rings are free*, Dokl. Akad. Nauk SSSR **229** (1976), no. 5, 1063–1066. 469905
- [Swa67] R. G. SWAN: *The number of generators of a module*, Math. Z. **102** (1967), 318–322. 218347

DEPARTMENT OF MATHEMATICS, PURDUE UNIVERSITY, WEST LAFAYETTE, IN, USA
Email address: `ma326@purdue.edu`